Dosely

A Smart Medication Analysis System Using AI and Patient Medical History

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Software Requirement Specification

Purpose Of The System

The Medicine Scanner Application is designed to help users determine whether a specific medicine is safe for them to use based on their personal and health-related information. The system allows users to scan a medicine using the mobile phone camera, select an image from the phone gallery, or search for a medicine by name.

The main purpose of the system is to reduce the risk of taking unsafe medicines by providing personalized safety checks, clear warnings, and explanations. The system also helps users manage their medicines by offering a scheduling feature with reminders and conflict prevention. The application aims to support safer medicine usage but does not replace professional medical advice.

Scope Of The System

In Scope

The following functionalities are included within the scope of the Medicine Scanner Application:

- Users can create an account, log in, and reset their password.
- Users must answer mandatory health-related and personalization questions before using the system.
- Users can scan medicines using the phone camera or select an image from the phone gallery.

- Users can search for medicines by entering the medicine name manually.
- The system checks medicine safety based on the user's personalized health profile.
- The system displays clear results indicating whether a medicine is safe or not safe, along with reasons and medicine descriptions.
- Users can add medicines to a schedule, update dosages and times, remove medicines, and receive reminders.
- The system prevents medicine conflicts based on scheduled medicines and user health information.
- Users can access a home page that provides navigation to all system features.
- Users can view health-related articles and news in the library and read full articles using a "Read More" option.
- Users can access a chatbot for guidance and general assistance.
- Users can manage settings such as language selection, profile editing, personalization updates, FAQ, About Us, Reach Us, and community links.
- The system requires users to update personalization information every three months to continue using the application.
- The application supports both Arabic and English languages.
- The system operates as a mobile application on Android and iOS platforms

Out Of Scope

The following functionalities are not included within the scope of the system:

- The system does not provide medical diagnosis or treatment.
- The system does not replace doctors, pharmacists, or professional healthcare services.
- The system does not prescribe medicines or recommend medical treatments.
- The system does not guarantee medical accuracy equal to hospitals or clinical systems.
- The system does not support offline usage; an internet connection is required.

- The system does not conduct live consultations with medical professionals.
- The system does not manage hospital records or official medical histories.

Product Perspective

The Medicine Scanner Application is a standalone mobile application developed for Android and iOS platforms. The system is designed as a user-centered health support tool that focuses on personalized medicine safety and medicine management.

Users interact with the system through a simple and intuitive mobile interface that includes scanning, searching, scheduling, and personalization features. The system relies on user-provided health information and external medicine databases to analyze medicine safety. All processing and validation are performed online, and the system enforces strict rules to ensure that user information is kept up to date. The application is designed to be easy to use, secure, and supportive, while clearly stating that it does not replace professional medical advice.

Product Requirement

Functional Requirement

Requirement ID	SRS-MED-001
Title	Scan Medicine using phone camera
Priority	High Priority
Description	The system shall allow the user to scan a medicine by using the mobile phone camera. The system shall capture the medicine box text and extract relevant information such as the medicine name, dosage, manufacturer, and other visible details using text recognition
Version	1.0

Requirement ID	SRS-MED-002
Title	Scan Medicine from Image Gallery
Priority	High Priority
Description	The system shall allow the user to select an existing image of a medicine from the mobile phone’s image gallery. The system shall extract the

	medicine information from the selected image using text recognition and use it to identify the medicine. This option shall be available as an alternative to scanning the medicine using the phone camera.
Version	1.0

Requirement ID	SRS-MED-003
Title	Search For a medicine
Priority	High Priority
Description	The system shall allow the user to manually enter the name of a medicine in a search field. The system shall search the medicine database using the entered medicine name and display the matching medicine results to the user. If no matching medicine is found, the system shall inform the user that the medicine is not available in the system. + it shows the history searches
Version	1.0

Requirement ID	SRS-ADD-001
Title	Add Medicine to Schedule
Priority	High Priority
Description	The system shall allow the user to add a medicine to the medicine schedule. The user shall be able to add a medicine directly after scanning a medicine or searching for a medicine by name using an “Add to Schedule” button. The system shall store the medicine name, dosage, and scheduled time in the schedule table.
Version	1.0

Functions related to the schedual

Requirement ID	SRS-SCH-001
Title	View Medicine Schedule
Priority	High Priority
Description	The system shall provide the user with a medicine schedule table that displays all medicines added by the user along with their dosage and scheduled times.

Version	1.0

Requirement ID	SRS-SCH-002
Title	Update Scheduled Medicine Dosage
Priority	High Priority
Description	The system shall allow the user to update an existing medicine in the medicine schedule by selecting an edit option represented by a pen icon. Upon entering the edit mode, the system shall display the list of scheduled medicines along with their names. The user shall be allowed to modify only the medicine dosage and scheduled time. Before saving the changes, the system shall verify that the updated information does not cause any conflicts with other scheduled medicines or the user's health information. If a conflict is detected, the system shall prevent the update and display a warning message. The edit mode shall also provide options to add or remove medicines from the schedule.
Version	1.0

Requirement ID	SRS-SCH-003
Title	Remove Medicine from the edit list
Priority	High Priority
Description	The system shall allow the user to remove a medicine from the medicine schedule. Once removed, the system shall stop all future notifications related to the removed medicine.
Version	1.0

Requirement ID	SRS-SCH-004
Title	notify Medicine from the edit list
Priority	High Priority
Description	The system shall notify the user when it is time to take a scheduled medicine based on the defined dosage time. Notifications shall be sent through the mobile application to remind the user to take the medicine.
Version	1.0

Requirement ID	SRS-SCH-005
Title	Prevent Medicine Schedule Conflicts
Priority	High Priority
Description	The system shall check for conflicts when the user attempts to add a new medicine to the schedule. If a conflict is detected between the new medicine and existing scheduled medicines based on user health information or medicine interactions, the system shall prevent the addition and display a warning message to the user explaining the conflict.
Version	1.0

Requirement ID	SRS-HOM-001
Title	Home Page Navigation
Priority	High Priority
Description	The system shall provide a home page that serves as the main entry point for the user after logging in. The home page shall display and provide access to all main system functionalities, including scanning a medicine, searching for a medicine, viewing the medicine schedule, and accessing the medicine library. The home page shall allow users to easily navigate between different features of the application.
Version	1.0

Requirement ID	SRS-LIB-001
Title	library
Priority	High Priority
Description	The system shall provide a medicine library that contains health-related news, articles, and educational content. The library shall display a list of articles with a brief summary for each item. Users shall be able to browse the library content from the application.
Version	1.0

Requirement ID	SRS-LIB-002

Title	Read More Button
Priority	High Priority
Description	The system shall allow the user to view the full content of an article by selecting a “Read More” option from the medicine library. Upon selection, the system shall display the complete article details, including the full text and any additional related information.
Version	1.0

Loggin /create an acc/ forget pass /

Requirement ID	SRS-AUTH-001
Title	Create Account
Priority	
Description	The system shall allow a new user to create an account by entering required personal information, including name, email address, and password, phone number.The system shall validate the entered information and prevent account creation if mandatory fields are missing or invalid.
Version	1.0

Requirement ID	SRS-AUTH-002
Title	Answer Health-Related Questions questions from (personalized)
Priority	
Description	The system shall require the user to answer a set of predefined health-related questions, These questions shall include lifestyle habits, allergies, current medications and supplements, medical conditions, and reproductive health information. These questions shall be used to personalize the system and determine medicine suitability for the user. The system shall not allow the user to access any system functionality until all required questions have been answered.
Version	1.0

Requirement ID	SRS-AUTH-003
Title	User Login
Priority	

Description	The system shall allow registered users to log in by entering their email address and password. The system shall authenticate the user credentials and grant access to the application upon successful verification. If the entered credentials are incorrect, the system shall display an appropriate error message.
Version	1.0

Requirement ID	SRS-AUTH-004
Title	Forgot Password
Priority	
Description	The system shall provide a “Forgot Password” option that allows users to reset their password. The system shall verify the user’s identity using the registered email address and allow the user to create a new password after successful verification.
Version	1.0

settings

Requirement ID	SRS-SET-001
Title	View and Edit User Profile
Priority	
Description	The system shall allow the user to view and edit their profile information from the settings section. The profile shall display the user’s name and profile photo. The user shall be able to upload or change their profile photo.
Version	1.0

Requirement ID	SRS-SET-002
Title	Personalized Health Information
Priority	

Description	The system shall require the user to provide detailed personalized health information through a set of mandatory questions. These questions shall include lifestyle habits, allergies, current medications and supplements, medical conditions, and reproductive health information. The system shall ensure that all required questions are answered before allowing the user to complete account creation or access any system functionality.
Version	1.0

Requirement ID	SRS-SET-003
Title	Periodic Personalization Update
Priority	
Description	The system shall require the user to update their personalized health information every three months. Upon reaching the update period, the system shall notify the user and restrict access to all application functionalities until the required information has been reviewed and updated. The system shall allow access only after the update process is successfully completed.
Version	1.0

Requirement ID	SRS-SET-004
Title	Restrict Access for Outdated Personalization Information
Priority	
Description	The system shall monitor the validity period of the user’s personalized health information. If the information exceeds the allowed update period, the system shall block access to all system features and redirect the user to the personalization update screen until the information is updated
Version	1.0

Requirement ID	SRS-SET-005
Title	Change Application Language
Priority	
Description	The system shall allow the user to change the application language from the settings section. The system shall support both Arabic and English languages. Once the language is changed, the system shall apply the selected language across all application screens.

Version	1.0

Requirement ID	SRS-SET-006
Title	Share Application
Priority	
Description	The system shall allow the user to share the application with others through available sharing options on the mobile device. The system shall generate a shareable link that directs users to download the application.
Version	1.0

Requirement ID	SRS-SET-007
Title	View About Us Information
Priority	
Description	The system shall provide an “About Us” section that displays general information about the application, its purpose, and its goals. This section shall be accessible from the settings menu.
Version	1.0

Requirement ID	SRS-SET-008
Title	Reach Us
Priority	
Description	The system shall provide a “Reach Us” option that allows users to view contact information or communication channels for application support. This section shall enable users to contact the support team for inquiries or assistance.
Version	1.0

Requirement ID	SRS-SET-009

Title	View Frequently Asked Questions
Priority	
Description	The system shall provide a Frequently Asked Questions (FAQ) section that displays common questions and answers related to application usage, medicines, and general guidance. The FAQ section shall be accessible from the settings menu.
Version	1.0

Requirement ID	SRS-SET-011
Title	Access Discord Community
Priority	
Description	The system shall provide a link that allows users to access the application's Discord community. Selecting the link shall redirect the user to the official Discord server using an external browser or application.
Version	1.0

chatbot

Requirement ID	SRS-BOT-001
Title	Chatbot Assistance
Priority	
Description	The system shall allow the user to interact with a chatbot to ask questions related to medicines, application usage, and general health information. The chatbot shall provide automated responses to guide the user and offer basic assistance. The chatbot shall not replace professional medical advice.
Version	1.0

Non-Functional Requirement

Requirement ID	SRS-NFR-001
Title	System Performance
Priority	High Priority
Description	The system shall provide fast and responsive performance. Medicine scanning and text recognition shall be completed within 3 seconds under normal conditions. Medicine search results shall be displayed within 2 seconds, and navigation between application screens shall not exceed 1 second.
Version	1.0

Requirement ID	SRS-NFR-002
Title	System Availability
Priority	High Priority
Description	The system shall be available 24/7, except during scheduled maintenance. The system shall maintain a minimum uptime of 99% to ensure continuous access for users.
Version	1.0

Requirement ID	SRS-NFR-003
Title	Security and Data Privacy
Priority	High Priority
Description	The system shall protect user data by encrypting passwords and securely storing personal and health information. Unauthorized access to sensitive data shall be prevented, and privacy regulations shall be respected.
Version	1.0

Requirement ID	SRS-NFR-004
Title	Usability
Priority	High Priority
Description	The system shall provide an intuitive and user-friendly interface suitable for

	users with different technical backgrounds. All system features shall be easy to understand without requiring training.
Version	1.0

Requirement ID	SRS-NFR-005
Title	Reliability
Priority	High Priority
Description	The system shall operate reliably with minimal failures. In case of system errors, the system shall recover gracefully without data loss and notify the user appropriately.
Version	1.0

Requirement ID	SRS-NFR-006
Title	Scalability
Priority	High Priority
Description	The system shall support an increasing number of users and stored medicines without performance degradation. The system architecture shall allow future feature expansion.
Version	1.0

Requirement ID	SRS-NFR-007
Title	Compatibility
Priority	High Priority
Description	The system shall be compatible with modern mobile devices and different screen sizes. The application shall function correctly across supported operating systems.
Version	1.0

Requirement ID	SRS-NFR-008

Title	Maintainability
Priority	Medium Priority
Description	The system shall be developed using a modular design to simplify maintenance, debugging, and future updates without affecting existing functionality.
Version	1.0

Requirement ID	SRS-NFR-009
Title	Localization and Language Support
Priority	High Priority
Description	The system shall support both Arabic and English languages. All application content shall update automatically based on the selected language.
Version	1.0

Requirement ID	SRS-NFR-010
Title	Legal and Ethical Compliance
Priority	High Priority
Description	The system shall provide health-related information for educational purposes only and shall not replace professional medical advice. Clear disclaimers shall be displayed to users.
Version	1.0

User Characteristics

The system is designed to support different categories of users with varying backgrounds, technical skills, and health needs. The primary user characteristics are described below.

1. General Users (Patients)

- Users may have basic to moderate technical skills and are expected to be familiar with using smartphones and mobile applications.
- Users may have limited medical knowledge; therefore, the system must present medicine information in a clear, simple, and understandable manner.
- Users may include individuals of different age groups, including young adults and elderly users.
- Users may have personalized health conditions, such as allergies, chronic diseases, or ongoing medications, which require accurate and safe system guidance.
- Users expect reminders, notifications, and alerts to be clear, timely, and easy to follow.

2. Health-Conscious Users

- These users actively monitor their health and medication schedules.
- They may frequently use features such as medicine scanning, medicine search, and schedule management.
- They expect accurate, up-to-date, and reliable health information from the system.
- They are likely to interact with the medicine library and educational content regularly.

3. Elderly Users

- Elderly users may have limited technical experience and slower interaction speed.
- The system should support simple navigation, large readable text, clear icons, and minimal interaction steps.
- Notifications and reminders should be easy to understand and noticeable.

4. Administrators / Support Team

- Administrators are expected to have advanced technical knowledge.
- They are responsible for managing system content such as medicine data, health articles, and application support.

- They require secure access and system monitoring capabilities.

General Constraints

The system shall be developed as a mobile application and its functionality may be limited by the hardware capabilities of the user's device, including camera quality and processing power. The system relies on image processing and text recognition technologies, and the accuracy of medicine scanning may be affected by image clarity, lighting conditions, and the quality of medicine packaging. Certain features of the system, such as medicine lookup, chatbot assistance, and content updates, require an active internet connection, and limited connectivity may restrict system functionality. The availability and accuracy of system results depend on the correctness and completeness of the medicine database. The system must comply with applicable data protection, security, and privacy regulations to protect user information. Additionally, the system is intended for informational and educational purposes only and does not replace professional medical consultation. The development of the system is constrained by project timelines, academic requirements, and available resources. Initially, the system shall support English and Arabic languages only, with the possibility of supporting additional languages in future versions.

Principle Stakeholder

User

Role: Primary system user
Tasks: Scan medicines, manage schedules, receive reminders, provide health information
Goals: Safe medication usage, avoid conflicts, and manage medicines easily

• System Developer

Role: Application designer and maintainer
Tasks: Develop, test, secure, and maintain the system
Goals: Deliver a reliable, secure, and scalable system

• Project Supervisor / Academic Evaluator

Role: Project oversight and evaluation
Tasks: Review documentation and system implementation
Goals: Ensure academic quality and requirement compliance

- Healthcare Information Provider

Role: Medical data source
Tasks: Provide and update medicine and health information
Goals: Ensure accuracy and safe medication guidance

SYSTEM COMPONENTS

Use Cases Description

1.Use Cases Related to Authentication & Account Management

A. Use Case 1:Create an account
   a. Primary Actor:User
   b. Precondition: The user is aware of what the system does
   c. Postconditions: the user has entered the informations correctly and the system sent the user to answer health related questions
   d. Flow of event (Main Scenario):
      i. The user opens the application after it downloads it on his phone
      ii. The user will have a choice to either create an account or Log in , and he presses the (Create an account) button
      iii. The user will fill all required fields with the information needed and presses the (Create an account) button
      iv. The system will successfully move to the next step
   e. Exceptions (Alternate Scenario):
      i. The user enters an (already available email or phone) or there is a missing field
         1. The system will show a message to inform the user that the email/ phone number is taken and highlights the field that needs a change
   f. Quality Requirement
   g. Priority Level
   h. Related functional requirement IDs: **SRS-AUTH-001**

B. Use Case 2: User Log in
   a. Primary Actor:User
   b. Precondition: The user is aware of what the system does
   c. Postconditions: the user has entered the informations correctly and the system sent the user to answer health related questions (only if the informations needed some updates),otherwise it successfully logs the user to his account
   d. Flow of event (Main Scenario):
      i. The user opens the application
      ii. The user will have a choice to either create an account or Log in , and he presses the (Log in) button
      iii. The user will fill all required fields with the information needed and presses the (Log in) button
      iv. The system will successfully move to the next step, either by taking him to update his health informations if it needs to be updated or successfully logs the user to his account

e. Quality Requirement
f. Priority Level
g. Related functional requirement IDs: **SRS-AUTH-003**

C. Use Case 3:Forgot Password
   a. Primary Actor: User
   b. Precondition: The user already has a registered account. The user is on the login screen.The user has access to the registered email (to receive the reset verification)
   c. Postcondition: The user successfully resets their password. The user can log in using the new password.
   d. Flow of Event:
      i. The user opens the login screen.
      ii. The user selects **"Forgot Password"**.
      iii. The system asks the user to enter their registered email address.
      iv. The user enters the email and submits the request.
      v. The system verifies that the email is registered in the system.
      vi. The system sends a password reset link or verification code to the user's email.
      vii. The user opens the email and uses the link/code.
      viii. The system asks the user to enter a new password and confirm it.
      ix. The user enters and confirms the new password.
      x. The system validates the new password rules (example: minimum length).
      xi. The system saves the new password and shows a success message.
      xii. The user returns to the login screen and logs in with the new password.
   e. Quality Requirement
   f. Priority Level
   g. Related Functional Requirement ID: SRS-AUTH-004

2.Use Cases Related to Personalization & Health Profile

A. Use Case 1:Answer health related Questions (Personalized questions)
   a. Primary Actor: User
   b. Precondition: The user has started registration OR the system asked the user to complete personalization before using the app.
   c. Postcondition: The user's answers are saved successfully. And the user becomes eligible to access the app features only if all required questions are completed.
   d. Flow of Event:
      i. The system displays a set of required personalization questions.

        ii. The user answers all questions (lifestyle habits, allergies, medications/supplements, medical conditions, reproductive health questions, etc.).
        iii. The user submits the answers.
        iv. The system validates that all required questions are answered (no missing fields).
        v. The system saves the answers to the user profile.
        vi. The system confirms completion and allows the user to proceed.
    e. Quality Requirement
    f. Priority Level
    g. Related Functional Requirement ID: SRS-AUTH-002, SRS-SET-002, SRS-SET-003

B. Use Case 2:Personalized health information
    a. Primary Actor: User
    b. Precondition: The user is logged in. and the user has access to Settings → Personalization.
    c. Postcondition:The user's personalized health profile is stored/updated successfully. And The system can use this profile later to check medicine safety.
    d. Flow of Event:
        i. The user opens Settings.
        ii. The user selects Personalization.
        iii. The system displays the user's saved health information (if it exists).
        iv. The user reviews the information and enters any missing details (allergies, medications/supplements, conditions, lifestyle habits, reproductive health).
        v. The user clicks Save/Submit.
        vi. The system validates the information (required fields are complete).
        vii. The system stores the information and confirms success.

    e. Quality Requirement
    f. Priority Level
    g. Related Functional Requirement ID: SRS-SET-002, SRS-SET-003

C. Use case 3:Periodic Personalization Update
    a. Primary Actor:User
    b. Precondition:The user is logged in OR tries to use the app after the 3-month period ends. The system detects that 3 months have passed since the last update.
    c. Postcondition:The user updates (or confirms) their personalization information. The system records the new "last updated" date and restores access to features.
    d. Flow of Event:
        i. The system detects that the user's personalization info is older than 3 months.
        ii. The system displays a message telling the user they must update their info.

iii. The system redirects the user to the **Personalization Update** screen.
iv. The user reviews the current information.
v. The user updates any changes (for example: new allergy, new medicine, pregnancy status, etc.).
vi. The user submits the updated information.
vii. The system validates and saves the updates.
viii. The system confirms completion and allows the user to continue using the app.

e. Quality Requirement
f. Priority Level
g. Related Functional Requirement ID: SRS-SET-004

D. Use Case 4:Restrict Access for outdated personalization information

a. Primary Actor: user
b. Precondition: The user attempts to use any feature (scan/search/schedule/library/chatbot). The system detects personalization information is outdated (older than 3 months).
c. Postcondition:The system blocks access to other app features. The user is redirected to update personalization information.
d. Flow of Event:
   i. The user opens the app and logs in (or returns to the app).
   ii. The user selects any feature from the home page.
   iii. The system checks the “last updated” date of personalization info.
   iv. If the info is outdated, the system blocks access.
   v. The system shows a warning message explaining the reason (info must be updated every 3 months).
   vi. The system redirects the user to the personalization update screen.
   vii. The user updates and submits the info.
   viii. After successful update, the system allows access to features again.
e. Quality Requirement
f. Priority Level
g. Related Functional Requirement ID: SRS-SET-005, SRS-SET-004

3. Use Cases Related to User Profile Management

A. Use Case 1: View and Edit User Profile

a. Primary Actor: User

b. Precondition:
The user is logged in to the system and has an existing user profile.

c. Postconditions:
The user profile information is updated successfully, or remains unchanged if an error occurs.

d. Flow of event (Main Scenario):
i. The user opens the application.
ii. The user navigates to the Settings page.
iii. The user selects the Profile option.
iv. The system displays the user's profile information.
v. The user selects the Edit option.
vi. The user modifies the profile information (name or profile photo).
vii. The user presses the Save button.
viii. The system validates and saves the updated information successfully.

e. Exceptions (Alternate Scenario):
i. The user enters invalid information or leaves a required field empty.

1. The system displays an error message and highlights the incorrect field.
   ii. The user session expires.

2. The system redirects the user to the login page.

f. Quality Requirement: Usability, Security

g. Priority Level: High

h. Related functional requirement IDs: SRS-SET-001

B. Use Case 2: Change Application Language

a. Primary Actor: User

b. Precondition:
The user is logged in and the system supports multiple languages.

c. Postconditions:
The selected language is applied across the application and saved as the default language.

d. Flow of event (Main Scenario):
i. The user opens the Settings page.
ii. The user selects the Language option.
iii. The system displays available languages (Arabic and English).
iv. The user selects a preferred language.
v. The system updates all application screens with the selected language.

e. Exceptions (Alternate Scenario):
i. Language resources fail to load.

1. The system displays an error message and keeps the previous language.

f. Quality Requirement: Usability, Compatibility

g. Priority Level: High

h. Related functional requirement IDs: SRS-SET-005

C. Use Case 3: View About Us Information

a. Primary Actor: User

b. Precondition:
The user can access the Settings page.

c. Postconditions:
The system displays the About Us information without modifying any data.

d. Flow of event (Main Scenario):
i. The user opens the Settings page.
ii. The user selects the About Us option.
iii. The system displays general information about the application.

e. Exceptions (Alternate Scenario):
i. The content is unavailable.

1. The system displays a message indicating that the information cannot be loaded.

f. Quality Requirement: Usability

g. Priority Level: Medium

h. Related functional requirement IDs: SRS-SET-007

D. Use Case 4: Reach Us

a. Primary Actor: User

b. Precondition:
The user can access the Settings page.

c. Postconditions:
The system displays contact information or opens an external communication channel.

d. Flow of event (Main Scenario):
i. The user opens the Settings page.
ii. The user selects the Reach Us option.
iii. The system displays available contact methods.
iv. The user selects a contact method.
v. The system opens the selected method using the device.

e. Exceptions (Alternate Scenario):
i. The device cannot open the selected contact method.

1. The system displays an error message.

f. Quality Requirement: Usability, Compatibility

g. Priority Level: Medium

h. Related functional requirement IDs: SRS-SET-008

E. Use Case 5: View FAQ

a. Primary Actor: User

b. Precondition:
The user can access the Settings page.

c. Postconditions:
The FAQ content is displayed successfully.

d. Flow of event (Main Scenario):
i. The user opens the Settings page.
ii. The user selects the FAQ option.
iii. The system displays a list of frequently asked questions.
iv. The user selects a question.
v. The system displays the corresponding answer.

e. Exceptions (Alternate Scenario):
i. FAQ content fails to load.

1. The system displays an error message.

f. Quality Requirement: Usability

g. Priority Level: Medium

h. Related functional requirement IDs: SRS-SET-009

F. Use Case 6: Access Discord Community

a. Primary Actor: User

b. Precondition:
The user can access the Settings page and an active Discord link exists.

c. Postconditions:
The system redirects the user to the Discord community page.

d. Flow of event (Main Scenario):
i. The user opens the Settings page.
ii. The user selects the Discord Community option.
iii. The system opens the Discord link in an external browser or application.

e. Exceptions (Alternate Scenario):
i. Discord application is not installed.

1. The system opens the link in a web browser.

f. Quality Requirement: Compatibility, Availability

g. Priority Level: Low

4. Use Cases Related to Medicine Identification & Lookup

A. Use Case 1: Scan Medicine Using Phone Camera

a. Primary Actor: User

b. Precondition:
The user is logged in to the system and has granted camera access permission.

c. Postconditions:
The medicine information is successfully identified and displayed, or the user is informed if the medicine cannot be identified.

d. Flow of event (Main Scenario):
i. The user opens the application.
ii. The user selects the Scan Medicine option from the home page.

iii. The system opens the phone camera interface.
iv. The user captures an image of the medicine box.
v. The system processes the image and extracts text information using text recognition.
vi. The system searches the medicine database using the extracted information.
vii. The system displays the identified medicine details to the user.

e. Exceptions (Alternate Scenario):
i. The camera permission is denied.

1. The system requests permission or instructs the user to enable it from settings.
   ii. The captured image is unclear or text recognition fails.

2. The system asks the user to retake the image.
   iii. The medicine is not found in the database.

3. The system displays a message indicating that the medicine is not available.

f. Quality Requirement: Performance, Reliability, Usability

g. Priority Level: High

h. Related functional requirement IDs: SRS-MED-001

B. Use Case 2: Scan Medicine From Image Gallery

a. Primary Actor: User

b. Precondition:
The user is logged in and has granted access to the image gallery.

c. Postconditions:
The medicine information is displayed successfully, or the system informs the user if identification fails.

d. Flow of event (Main Scenario):
i. The user opens the application.
ii. The user selects the Scan from Gallery option.
iii. The system opens the image gallery.
iv. The user selects an image of a medicine box.
v. The system extracts text from the selected image using text recognition.
vi. The system searches the medicine database using the extracted data.
vii. The system displays the medicine details to the user.

e. Exceptions (Alternate Scenario):
i. Gallery access permission is denied.

1. The system requests permission or shows an error message.
   ii. The selected image is unclear or unreadable.

2. The system asks the user to select another image.
   iii. The medicine is not found in the database.

3. The system displays a “medicine not available” message.

f. Quality Requirement: Reliability, Usability

g. Priority Level: High

h. Related functional requirement IDs: SRS-MED-002

C. Use Case 3: Search for the Medicine

a. Primary Actor: User

b. Precondition:
The user is logged in and has access to the search feature.

c. Postconditions:
The matching medicine results are displayed and the search is saved in the search history.

d. Flow of event (Main Scenario):
i. The user opens the application.
ii. The user navigates to the Search Medicine feature.
iii. The user enters the medicine name in the search field.
iv. The user presses the Search button.
v. The system searches the medicine database.
vi. The system displays a list of matching medicines.
vii. The user selects a medicine from the list.
viii. The system displays the full medicine details.

e. Exceptions (Alternate Scenario):
i. No matching medicine is found.

1. The system displays a message indicating that the medicine is not available.
   ii. The database or network connection is unavailable.

2. The system displays an error message and allows retry.

f. Quality Requirement: Performance, Availability, Usability

g. Priority Level: High

h. Related functional requirement IDs: SRS-MED-003

5. Use Cases Related to Medicine Schedule & Reminders

A. Use Case 1: View Medicine Schedule

a.Primary Actor: User

b.Precondition:

- The actor must be successfully logged into the system.
- The system must contain at least one recorded medication for the actor

c.Postcondition:

- The user has successfully viewed their daily or weekly medication plan.
- No changes are made to the database during this process (Read-only action).

d.Flow of Event:

1.Trigger: User navigates to the "Medication" or "My Schedule" tab from the main dashboard.

2.Request Data: The system retrieves the medication list associated with the User ID.

3.Display: The system displays the schedule, including:

Medication Name and Dosage (e.g., 500mg).

Specific Time Slots (e.g., 08:00 AM, After Lunch).

Special Instructions (e.g., "Take with food," "Avoid dairy").

Status indicator (e.g., Taken, Pending, Missed).

4.Termination: The user exits the schedule screen or selects a specific medicine for more details.

e.Quality Requirement :

1. Usability: The schedule must be easy to read for users with visual impairments (high contrast, adjustable font sizes).

2.Performance: The schedule should load in under 2 seconds.

3.Accuracy: The data must be synchronized in real-time with the central prescription database.

f.Priority Leve: **High**

g.Related Functional Requirement ID: **SRS-SCH-001**

B. Use Case 2: Add Medicine to the schedule

a.Primary Actor: User
b.Precondition :
- The user must be successfully logged into the system
- The user has already identified a medicine through scanning or searching

c.Postcondition
- The medicine name, dosage, and scheduled time are successfully stored in the schedule table
- Future notifications are enabled based on the set dosage time

d.Flow of Event:
1. **Trigger:** The user selects the "**Add to Schedule**" button directly after scanning a medicine box or searching for a medicine by name.
2. **Input Details:** The user enters the required dosage and selects the preferred scheduled times.
3. **Conflict Check:** The system automatically checks for conflicts between the new medicine and existing scheduled medicines based on the user's health profile (e.g., allergies or interactions).
4. **Verification:** If no conflicts are detected, the system saves the record.
5. **Confirmation:** The system displays a success message and updates the schedule view.

- *Exception (Conflict Detected):* If a conflict is found the system prevents the addition and displays a warning message explaining the risk.

e.Quality Requirement :
- **Performance:** Navigation between application screens shall not exceed 1 second.
- **Accuracy:** The system must accurately store the specific dosage and time to ensure user safety.
- **Usability:** All features must be easy to understand without requiring training.

f.Priority Level : High
g.Related Functional Requirement ID:
SRS-ADD-001: Add Medicine to Schedule.
SRS-SCH-005: Prevent Medicine Schedule Conflicts

C. Use Case 3: Update the scheduled Medicine dosage

a.Primary Actor: User
b.Precondition:
- The user must be successfully logged into the system.
- The system must contain at least one recorded medication for the actor.
- The user has navigated to the medicine schedule view.

c.Postcondition :

- The medicine dosage and/or scheduled time are successfully updated in the schedule table
- The system confirms the update and ensures no health or schedule conflicts exist.

d.Flow of Event:

1. **Select Edit:** The user selects an existing medicine in the schedule by clicking an edit option (represented by a pen icon)
2. **Enter Edit Mode:** The system displays the list of scheduled medicines along with their names.
3. **Modify Details:** The user modifies the medicine dosage and/or the scheduled time.
4. **Conflict Verification:** Before saving, the system verifies that the updated information does not cause conflicts with other scheduled medicines or the user's health profile (e.g., allergies or interactions).
5. **Save Changes:** If no conflicts are detected, the system saves the updated information.
6. **Error Handling:** If a conflict is detected, the system prevents the update and displays a warning message.
7. **Reliability:** In case of errors, the system shall recover gracefully without data loss

e.Quality Requirement

Performance: Navigation between application screens shall not exceed 1 second.

Usability: The interface must be intuitive and suitable for users with different technical backgrounds.

f.Priority Level : High

g.Related Functional Requirement ID: **SRS-SCH-002:** Update Scheduled Medicine Dosage

D. Use Case 4: Remove medicine from the edit list

a.Primary Actor : User

b.Precondition:

- The user is logged into the system
- The user has navigated to the medicine schedule view and entered "edit mode".

c.Postcondition:

- The medicine is removed from the schedule.
- All future notifications for that specific medicine are stopped.

d.Flow of Event:

1. The user enters edit mode (pen icon) on the schedule screen.
2. The user selects the medicine they wish to remove.
3. The user chooses the "Remove" option.
4. The system confirms the removal and updates the database

e.Quality Requirement :

- **Reliability:** The system shall recover gracefully without data loss if errors occur during deletion
- **Performance:** Navigation between application screens shall not exceed 1 second.

f.Priority Level: **High**

g.Related Functional Requirement ID: **SRS-SCH-003**.

E. Use Case 5: notify medicine

a.Primary Actor : User.

b.Precondition: A medicine has been successfully added to the schedule with a defined dosage time

c.Postcondition: The user is alerted to take their medication at the correct time.

d.Flow of Event:

1. The system monitors the current time against the scheduled dosage times.
2. When a match is found, the system triggers a push notification.
3. The mobile application displays the reminder on the user's device.

e.Quality Requirement:

Usability: Notifications should be easy to understand and noticeable, particularly for elderly users.

Availability: The system shall be available 24/7 to ensure timely reminders

f.Priority Level: High

g.Related Functional Requirement ID:**SRS-SCH-004**.

F. Use Case 6:Prevent Medicine Schedule Conflicts

a.Primary Actor: User

b.Precondition :

- The user attempts to add a new medicine or update an existing one.
- The user has completed their personalized health profile (allergies, current meds, etc.)

c.Postcondition

- The system either allows the schedule change or blocks it to prevent a health risk.

d.Flow of Event

1. The user initiates adding or updating a medicine
2. The system automatically compares the new medicine against the user's health information and existing medications
3. if a conflict (e.g., interaction or allergy) is detected:

- The system prevents the addition/update.
- A warning message is displayed explaining the specific conflict.

4.If no conflict is found, the change is saved

e.Quality Requirement

Accuracy: The system must accurately identify interactions to ensure user safety.

Performance: Conflict checks should be completed swiftly within the navigation time of under 1 second.

f.Priority Level : **High**

g.Related Functional Requirement ID: **SRS-SCH-005**

6.Use Cases Related to Home and Navigation

A. Use Case 1: Home Page Navigation

a. Primary Actor: User

b. Precondition:
The user is logged in successfully and the home page is displayed.

c. Postconditions:
The user is navigated to the selected feature page.

d. Flow of event (Main Scenario):
i. The user opens the application.
ii. The system displays the home page.
iii. The home page displays main navigation options such as scanning medicine, searching medicine, viewing the library, and accessing settings.
iv. The user selects one of the available options.
v. The system navigates the user to the selected feature page.

e. Exceptions (Alternate Scenario):
i. The selected feature fails to load.

1. The system displays an error message and allows the user to retry.

f. Quality Requirement: Usability, Performance

g. Priority Level: High

B. Use Case 2: Library

a. Primary Actor: User

b. Precondition:
The user is logged in and the library feature is available.

c. Postconditions:
The selected educational or health-related content is displayed to the user.

d. Flow of event (Main Scenario):
i. The user opens the application.
ii. The user navigates to the Library section from the home page.
iii. The system displays a list of available articles and resources.
iv. The user selects an article or resource.
v. The system displays the full content of the selected item.

e. Exceptions (Alternate Scenario):
i. Library content fails to load.

1. The system displays an error message and allows retry.

f. Quality Requirement: Usability, Availability

g. Priority Level: High

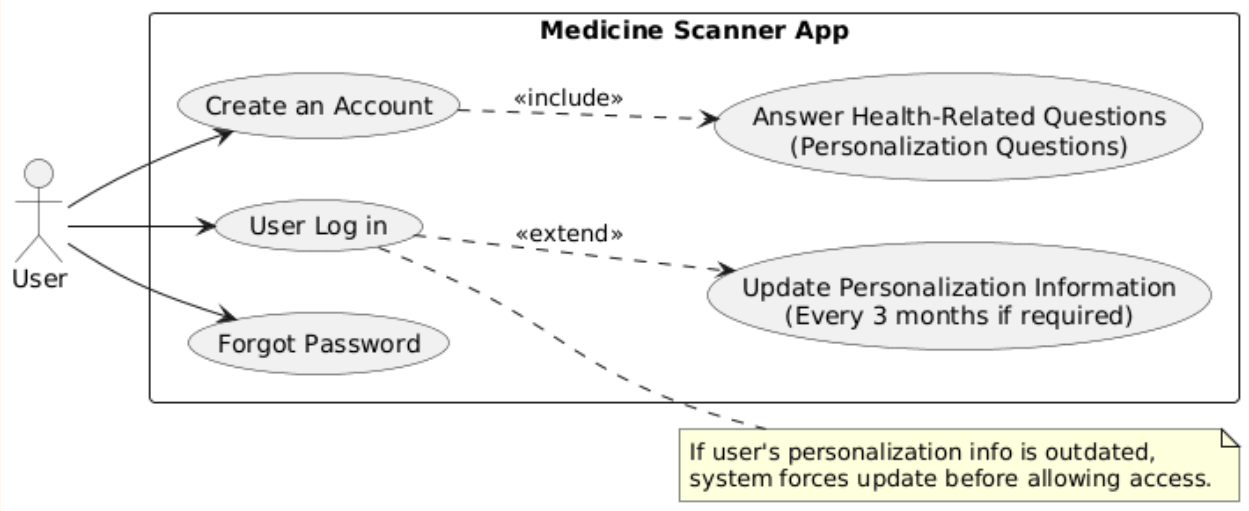
7. Use Cases Related to Chatbot Support

A. Use Case 1: Chat Bot Assistant
   a. Primary Actor: User
   b. Precondition:The user is logged in. The user has completed the required personalization questions (not blocked).
   c. Postcondition: The user receives a chatbot response to their question. The chat conversation is saved during the session
   d. Flow of event:
      i. The user opens the home screen.
      ii. The user taps the **Chatbot** button.
      iii. The system opens the chatbot interface.
      iv. The user types a question (for example: medicine usage, app help, general health guidance).
      v. The system sends the user's message to the chatbot service.
      vi. The chatbot analyzes the message and prepares an answer.
      vii. The system displays the chatbot's response to the user.
      viii. The user can continue asking more questions or exit the chatbot screen.
   e. Exceptions:
      i. Chatbot cannot answer / unclear question: The chatbot asks the user to clarify, or suggests FAQ/library articles.
      ii. Chatbot service unavailable: The system shows an error message and suggests using FAQ/Reach Us.
   f. Quality Requirement
   g. Priority Level

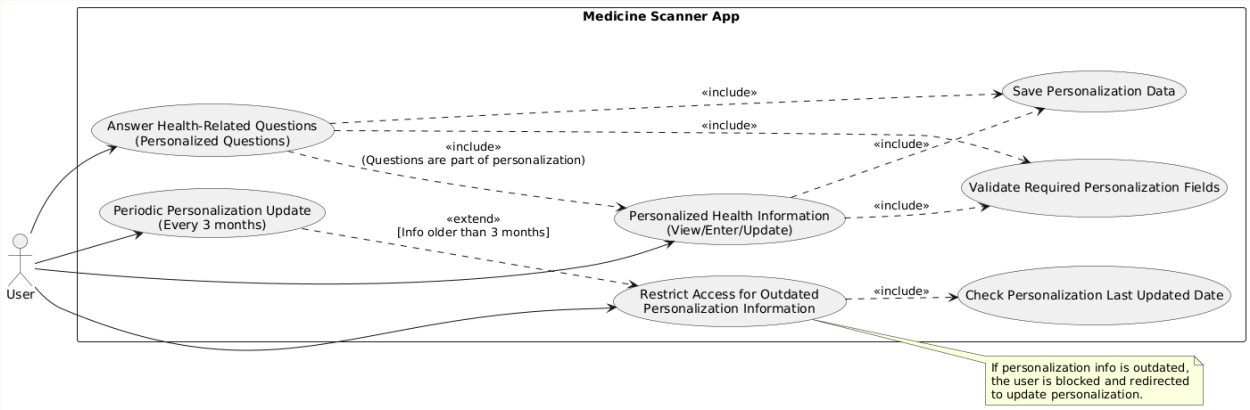
System Models

Functional Models (Use Cases)

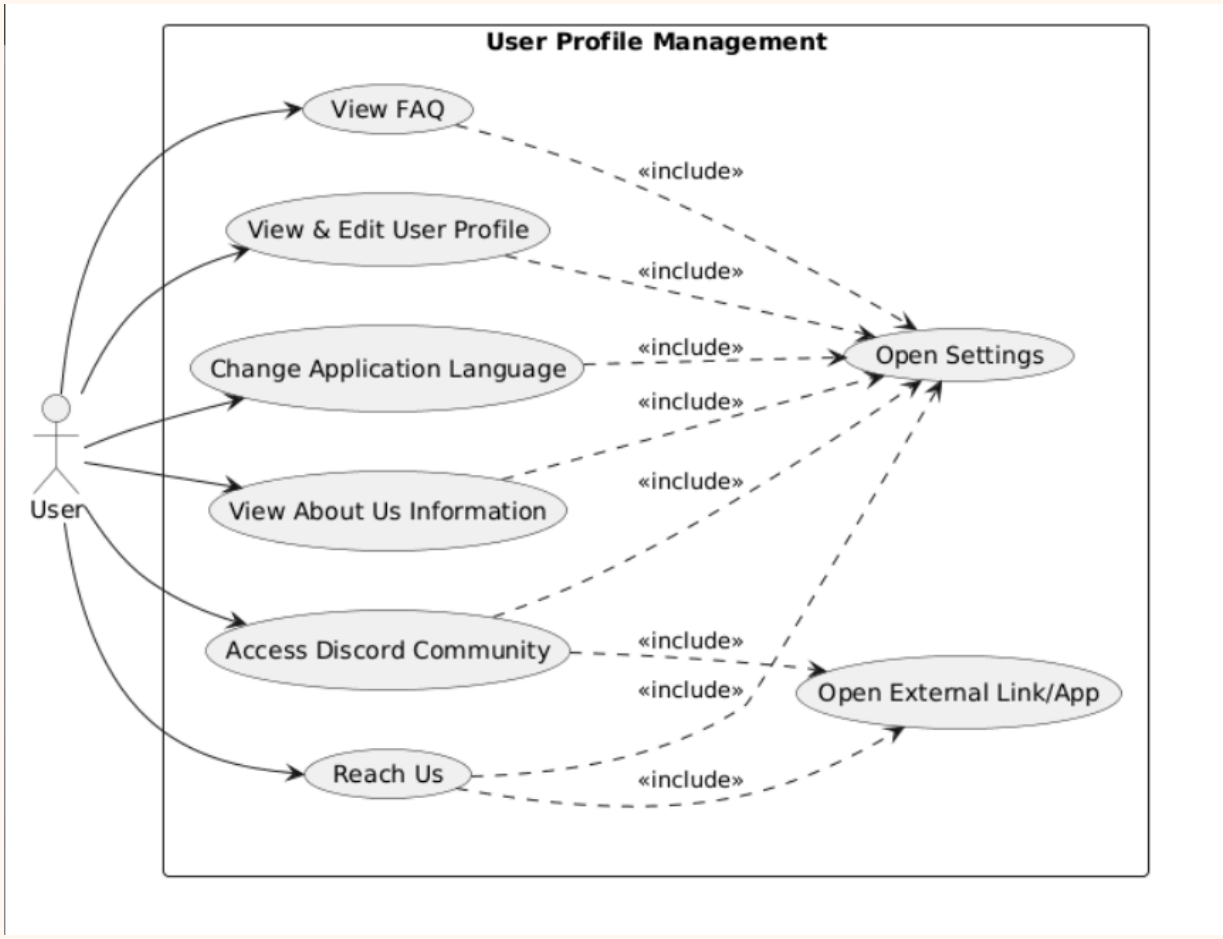
1.Use Cases Related to Authentication & Account Management



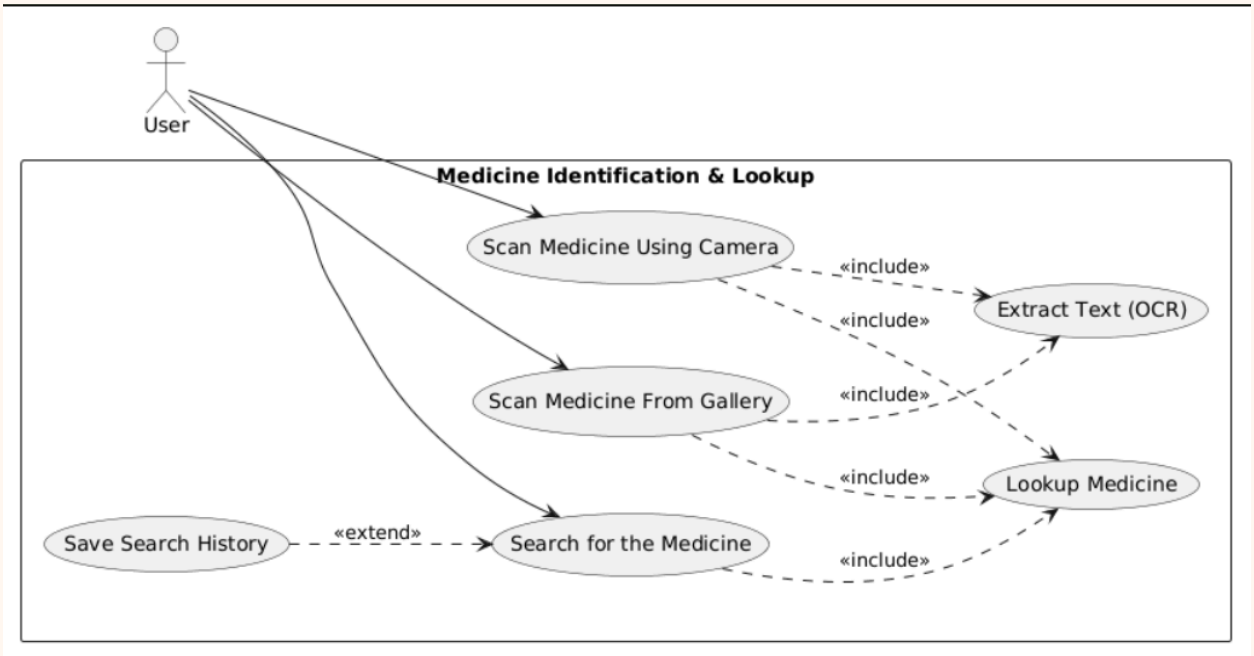
2.Use Cases Related to Personalization & Health Profile



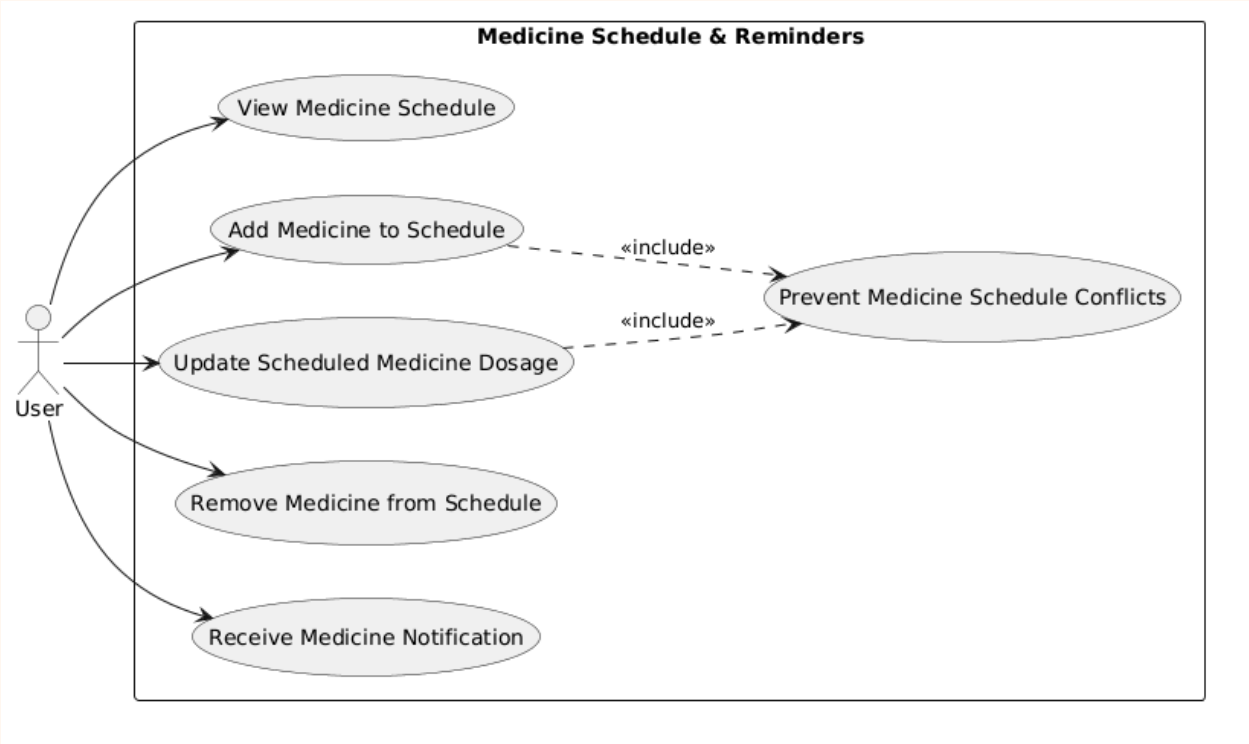
3. Use Cases Related to User Profile Management



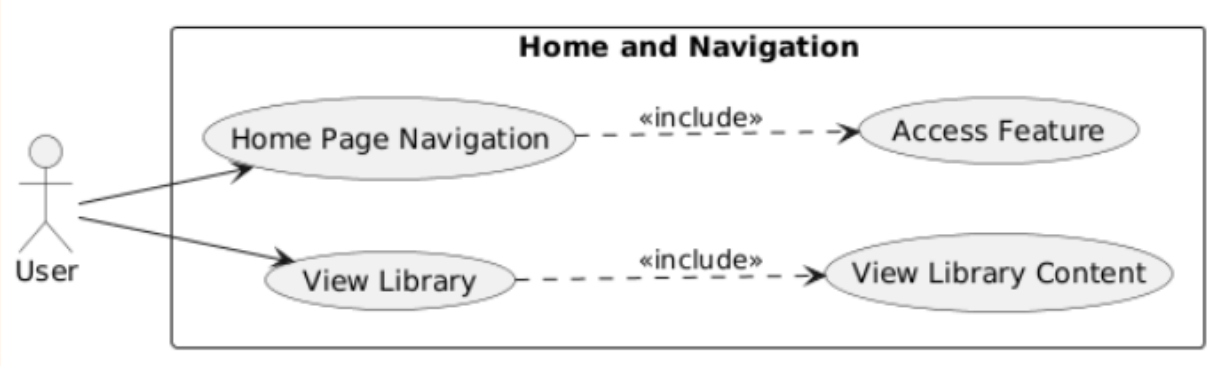
4. Use Cases Related to Medicine Identification & Lookup



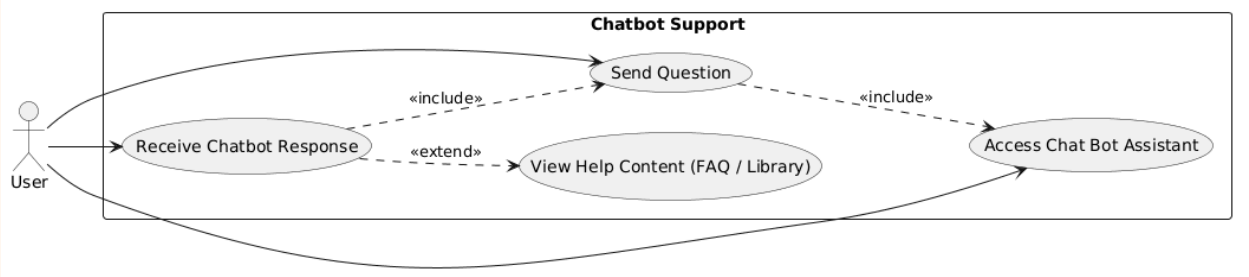
5.Use Cases Related to Medicine Schedule & Reminders



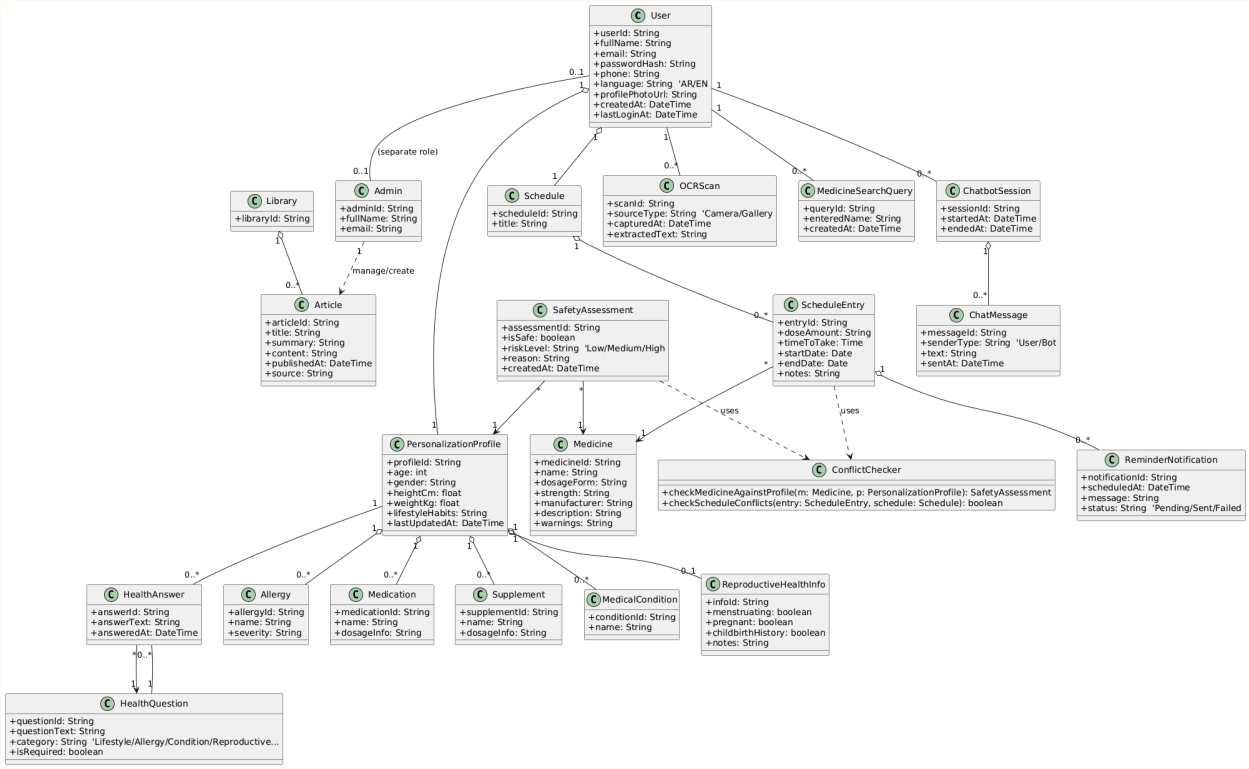
6.Use Cases Related to Home and Navigation



7.Use Cases Related to Chatbot Support



Object Models (UML Class Diagram)



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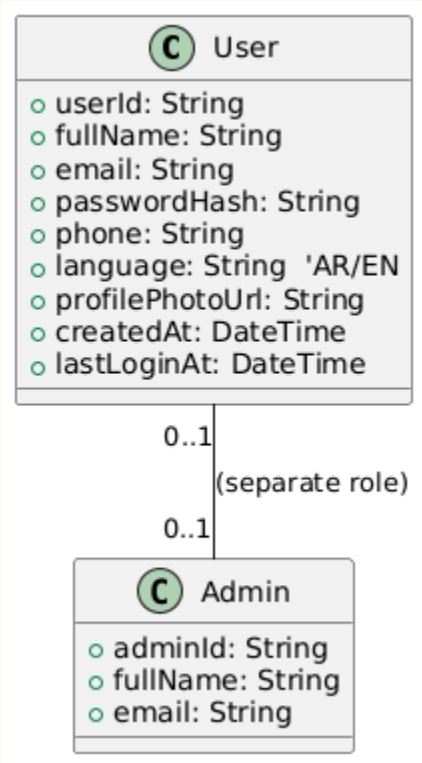
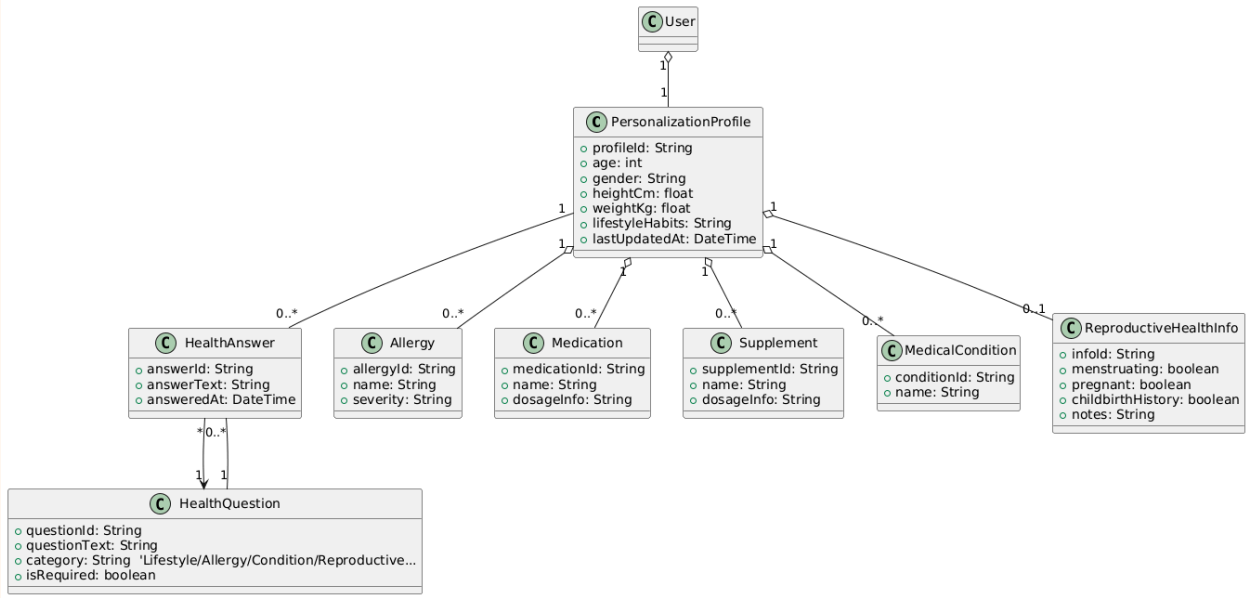


2.Use Cases Related to Personalization & Health Profile



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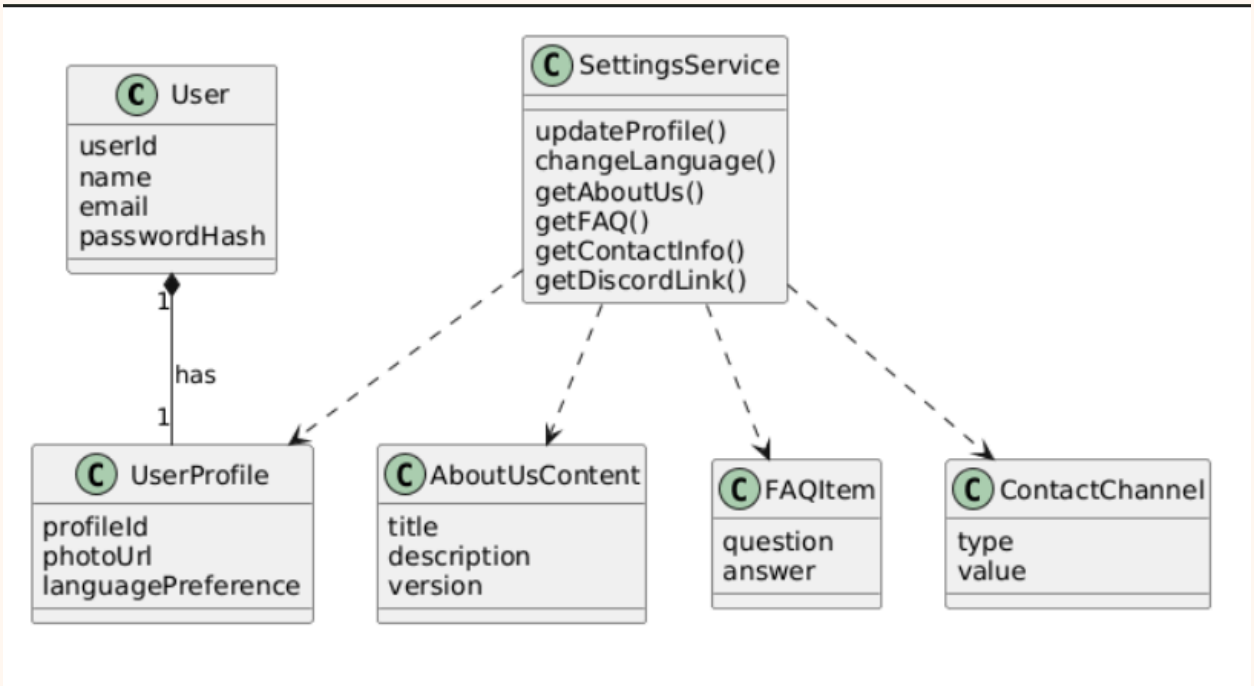
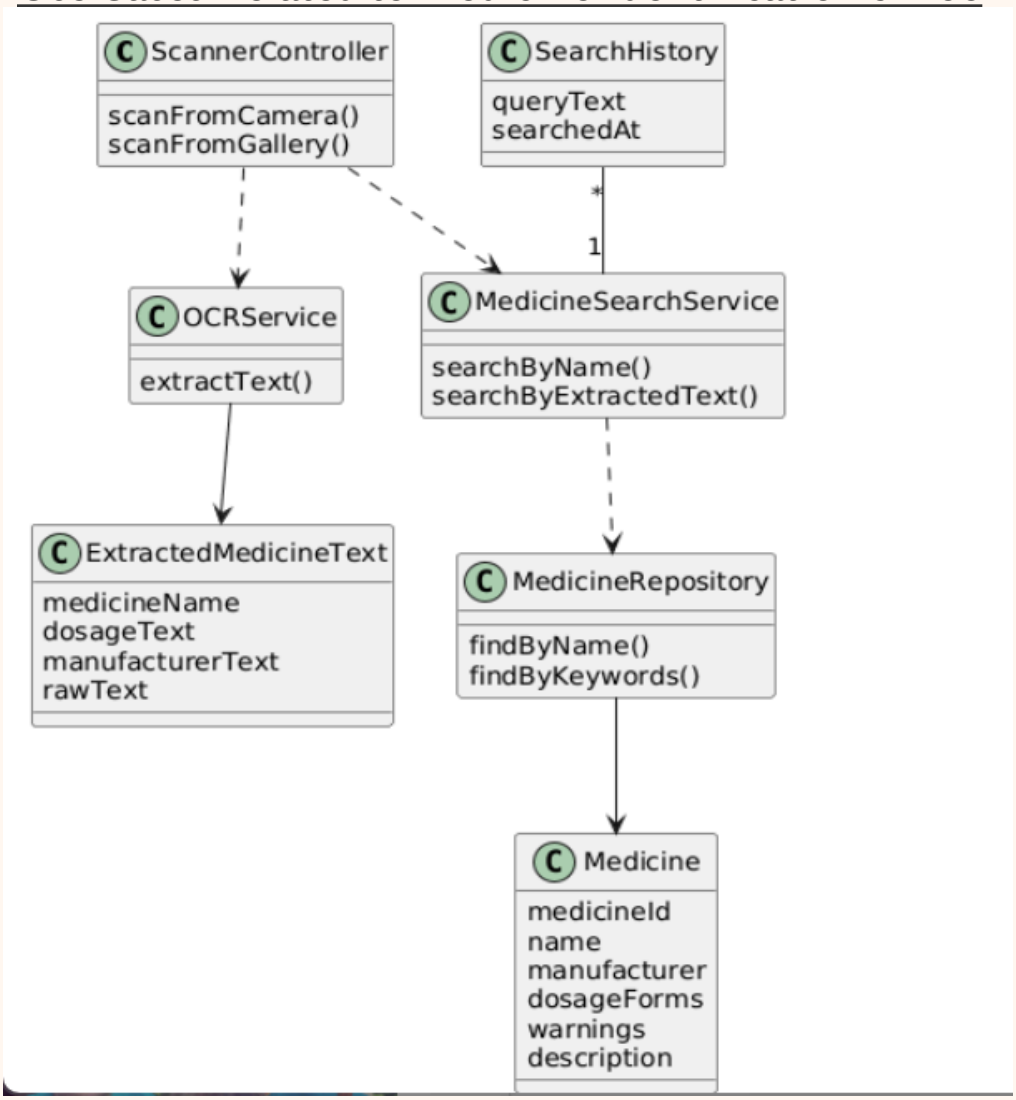
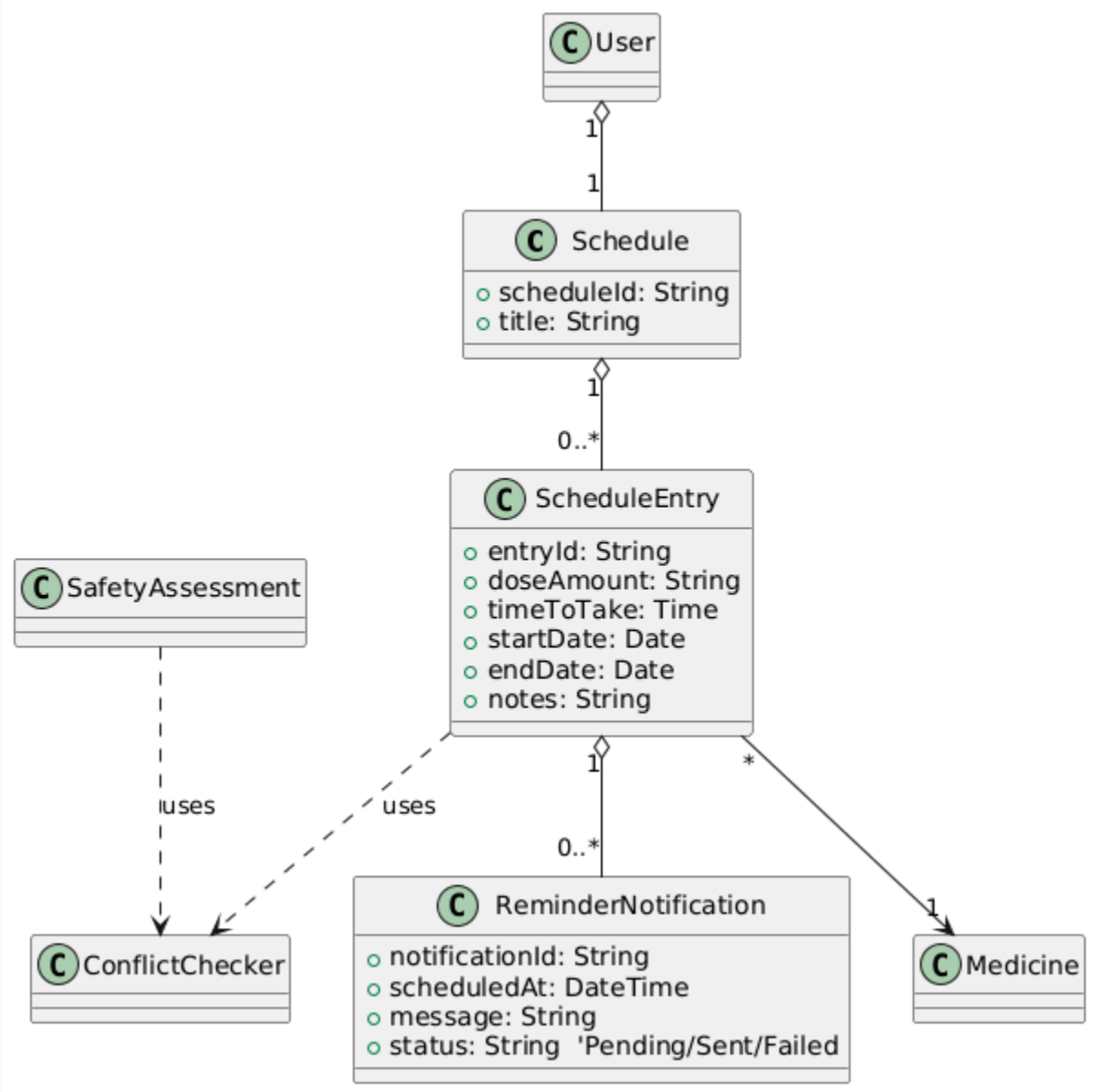


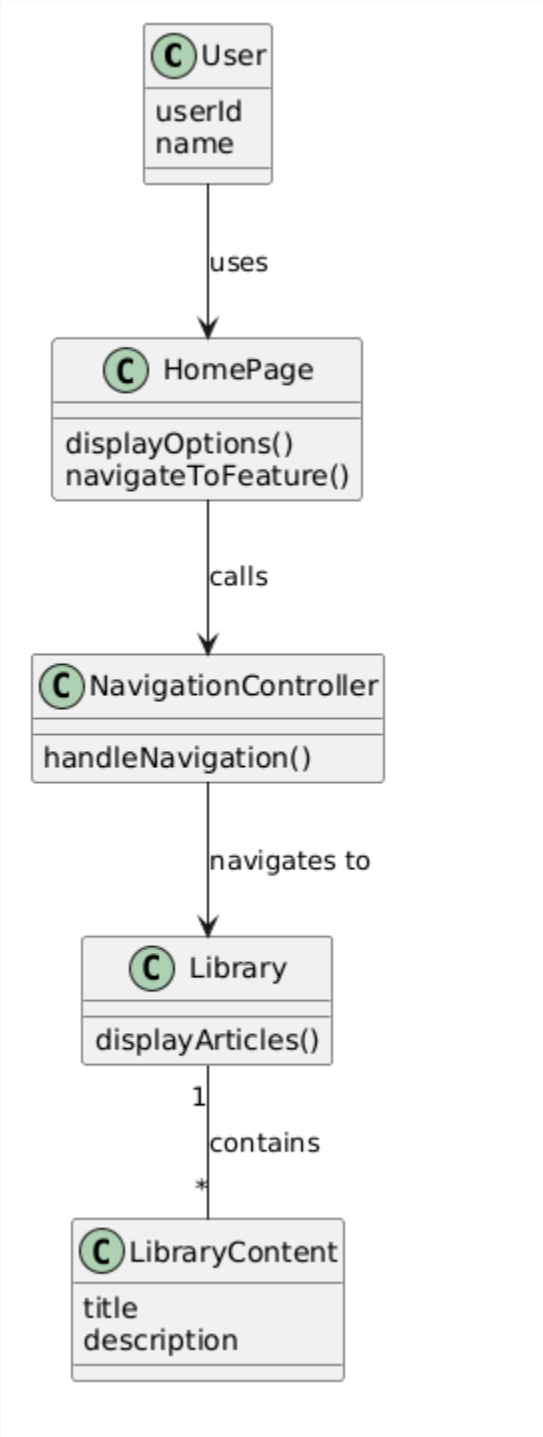
4. Use Cases Related to Medicine Identification & Loo



5.Use Cases Related to Medicine Schedule & Reminders



6. Use Cases Related to Home and Navigation



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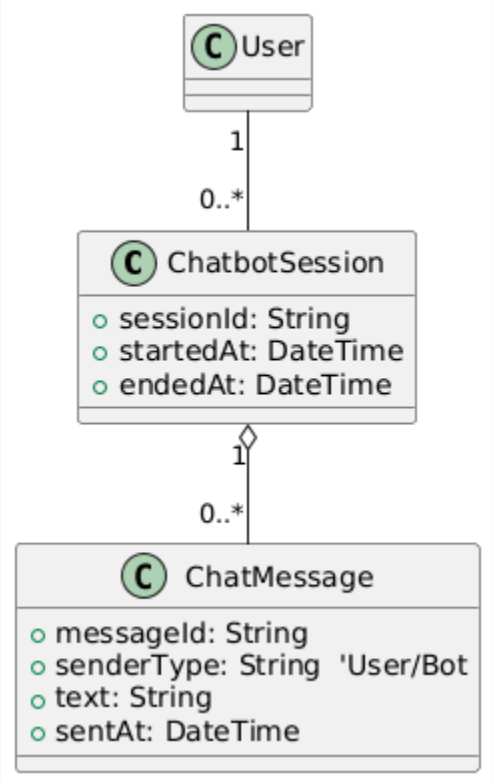

GUI Prototype